Ram Representations and Finite-Mode Certificates for Irrational-Triangle Billiards

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Abstract

This report examines the pinned manuscript claiming ergodicity of every nondegenerate triangle with an irrational angle. We separate its bounded-invariant-function argument from its exact flat-bundle geometry and classical spectral consequences. The entire single-paper source was read; relevant private Ram passages and current public primary literature were checked directly. The source's global theorem remains a source claim, not an independently established result of this report. We derive domain-correct representations, finite-mode holonomy residual certificates, a twisted-annulus coercivity inequality, and an explicit conditional lower bound for coefficient Rayleigh gaps. A finite-band transport estimate gives quantitative error budgets including the omitted derivative tail. We also formulate a source-dependent finite-cone-surface extension. None of these statements supplies regularity of arbitrary invariant indicators, a uniform infinite-mode gap, or an ergodic convergence rate. Boundary quantum-ergodicity normalization and real nodal-domain growth are checked conditional on classical ergodicity. Contribution assessment: C2/5, explicit local holonomy and finite-band certificates improve verification while the headline ergodicity claim and its low-regularity passage remain source-dependent.

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1 Problem, Status, and Scope

1.1 The pinned problem

Let $Q \subset \mathbb{R}^2$ be the interior of a fixed nondegenerate Euclidean triangle with angles $\alpha, \beta, \gamma \in (0, \pi)$ and $\alpha + \beta + \gamma = \pi$. Put $A_Q = \mathrm{Area}(Q) > 0$. A particle moves with unit speed, reflecting at each open side by $v \mapsto v - 2(v \cdot n)n$, for a unit side normal n. Remove trajectories hitting a vertex in either time direction. The probability measure on the resulting conull phase space is

$$d\mu = \frac{dA\, d\theta}{2\pi A_Q}. \tag{1}$$

Incoming and outgoing open-side states are identified by reflection when describing continuous trajectories; the boundary has zero flow measure. The manuscript [1], dated September 25, 2026, asserts

$$[\alpha/\pi \notin \mathbb{Q} \text{ or } \beta/\pi \notin \mathbb{Q} \text{ or } \gamma/\pi \notin \mathbb{Q}] \Longrightarrow (Q \times S^1, \mu, \Phi_t) \text{ is ergodic.} \tag{2}$$

Precisely, for every measurable E with $\mu(\Phi_t^{-1} E \triangle E) = 0$ for every $t \in \mathbb{R}$, the claim is $\mu(E) \in \{0, 1\}$. These equalities hold modulo null sets separately for each time. They do not assume a simultaneous pointwise representative invariant outside one common null set. No Diophantine or parameter-genericity hypothesis is imposed. Exactly one irrational angle and two rational angles are impossible since their ratios sum to one.

This is neither unique ergodicity nor weak mixing, mixing, an equidistribution claim for every orbit, or a rate. The catalogue family title mentions weak mixing; the assigned single paper actually claims *ergodicity*. This report analyzes that paper.

1.2 Evidence levels and assessment scales

The snapshot is identified by commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The complete 825-line `build/source/main.tex` and its bibliography were read. The local snapshot is not a Git worktree; the pin is a provenance identifier, not the result of a successful local `git rev-parse`. Input hashes and read ranges are retained in the dedicated evidence directory.

We distinguish source claims and their proof map, locally proved certificates, imported classical theorems with checked hypotheses, and conditional transfer of a specified source analytic package. An independent bounded reading of the energy and two-limit chain found no decisive gap. This is not external peer review, machine formal verification, or independent endorsement of the global theorem.

Applicability **L4/5** means substantial relevance of the private polygonal-flow material. Contribution **C2/5** assesses realized improvements relative to the pinned manuscript. The scales are distinct. Fourier decomposition, Poincare inequalities, unitary holonomy, physical Gram metrics, and quantum ergodicity are classical imported mathematics, not inventions of Ram Calculus.

2 Flat Double, Bundles, and Domains

2.1 Doubling and measure

Glue two closed copies of Q along corresponding sides, with opposite orientations, and delete their vertices. The connected oriented flat surface M is topologically a sphere with three punctures. Its compact completion has cone angles $2\alpha, 2\beta, 2\gamma$ and area $2A_Q$. Write V for the cone set and

$$dm = \frac{dA}{2A_Q}, \qquad d\nu = dm\,\frac{d\theta}{2\pi}.$$

The fold $SM \to Q \times S^1$ pushes ν to μ: interior points have two equally weighted lifts and collisions swap sheets. Ergodicity upstairs implies ergodicity downstairs; the reverse implication for an arbitrary finite extension is not asserted.

A fixed finite side word unfolds vertex hits into lines through a specified point, a null condition in position-direction space. The union over words, vertices, and time orientations is countable. Collision accumulation cannot occur away from vertices because open sides are regular upstairs. Near a vertex the unfolded wedge has a fixed positive angle; a line avoiding its tip sweeps at most π in polar angle, crossing finitely many sectors. This excludes another finite-time obstruction. The conull complete geodesic flow Ψ_t preserves ν by local translation. These facts check the setup, not ergodicity.

2.2 Coefficient transitions

In an oriented flat chart $z = x_1 + ix_2$ set

$$\partial = \tfrac{1}{2}(\partial_{x_1} - i\partial_{x_2}), \qquad \bar\partial = \tfrac{1}{2}(\partial_{x_1} + i\partial_{x_2}),$$

$$X = e^{i\theta}\partial + e^{-i\theta}\bar\partial, \quad Y = ie^{i\theta}\partial - ie^{-i\theta}\bar\partial, \quad \Theta = \partial_\theta. \tag{3}$$

Thus $X = \cos\theta\,\partial_{x_1} + \sin\theta\,\partial_{x_2}$, $Y = -\sin\theta\,\partial_{x_1} + \cos\theta\,\partial_{x_2}$ and $[X, Y] = 0$, $[\Theta, X] = Y$, $[\Theta, Y] = -X$ locally. For $f \in L^2(SM, \nu)$, let $P_j f = f_j(x)e^{ij\theta}$. Under $z' = e^{i\phi}z + c$, $\theta' = \theta + \phi$, so

$$f_j' = e^{-ij\phi}f_j, \quad (\partial f_j)' = e^{-i(j+1)\phi}\partial f_j, \quad (\bar\partial f_j)' = e^{-i(j-1)\phi}\bar\partial f_j. \tag{4}$$

Consequently f_j is a section of a flat unitary line bundle L_j, and $\partial : L_j \to L_{j+1}$, $\bar\partial : L_j \to L_{j-1}$. Treating every f_j as a globally single-valued scalar is generally wrong. Direct differentiation gives, also distributionally,

$$(Xf)_j = \partial f_{j-1} + \bar\partial f_{j+1}, \qquad (Yf)_j = i\partial f_{j-1} - i\bar\partial f_{j+1}. \tag{5}$$

The summands have the same target twist. Parseval supplies the unitary map

$$\mathcal{F} : L^2(SM,\nu) \longrightarrow \widehat{\bigoplus}_{j\in\mathbb{Z}} L^2(M, L_j; dm), \qquad \|f\|_2^2 = \sum_j \|f_j\|_2^2 . \tag{6}$$

Surjectivity follows by taking L^2 Fourier sums of square-summable sections; the transitions make the local sums agree. This specializes the classical framework of Forni–Moll [2].

2.3 Three different domains

First, $Xf = 0$ may mean a distributional equation tested on $C_c^\infty(SM)$. A bounded invariant indicator satisfies it, with no assumed transverse derivative. More precisely, the measure-preserving flow defines a strongly continuous unitary Koopman group on L^2, whose generator has domain

$$\mathrm{Dom}(\mathsf{X}) = \{f : \lim_{t\to 0}(f\circ\Psi_t - f)/t \text{ exists in } L^2\}.$$

Every invariant L^2 function belongs to this domain with derivative zero. That does not place it in a joint X, Y Sobolev domain. We do not identify every maximal distributional domain with the generator domain without a proof of that identification.

Second, a coefficient may lie in

$$\mathcal{W}_j = \{u \in L^2(M, L_j) : \nabla u \in L^2 \text{ weakly on the regular surface}\}.$$

This declares no cone-tip trace or self-adjoint realization. Finiteness of every $\|\nabla f_j\|_2$ does not imply finiteness of their squared sum.

Third, for the finite-band certificate choose the Friedrichs form domain

$$\mathcal{V}_j = \overline{C_c^\infty(M, L_j)}^{(\|u\|_2^2 + \|\nabla u\|_2^2)^{1/2}}, \qquad q_j(u) = \|\nabla u\|_2^2 . \tag{7}$$

The associated nonnegative self-adjoint Δ_j has domain consisting of $u \in \mathcal{V}_j$ such that $q_j(u, v) = \langle h, v\rangle$ for some $h \in L^2$ and every $v \in \mathcal{V}_j$. Inner products here are linear in the first argument. A horizontal form field has coefficients in $\mathcal{V}_j$ and a finite sum of the form norms.

Constants belong to $\mathcal{V}_0$: at a cone of angle 2α, cut off below ε^2, use an affine function of $\log r$ up to ε, then set the cutoff to one. Its energy is $2\alpha/|\log\varepsilon| \to 0$ and omitted area tends to zero. This classical zero-capacity calculation does not make discontinuous invariant indicators horizontally Sobolev.

For $u \in \mathcal{V}_j$, compact-core integration by parts and closure give

$$\|\partial u\|_2^2 = \left\|\bar\partial u\right\|_2^2 = \tfrac{1}{4} q_j(u). \tag{8}$$

On the core the difference of the first two squared norms is a multiple of the integral of $du \wedge d\overline{u}$. Its primitive $u\, d\overline{u}$ is global, because unitary transition factors cancel, and has compact support. Stokes followed by form approximation proves the identity. It is not claimed for unspecified singular extensions.

3 Source Proof Map and Analytic Boundary

This dependency map covers the entire single-paper source. Literal source labels identify the steps; this section is not an alternative proof of the low-regularity theorem.

Source location	Content and role
`sec:geometry`, 177–271	Conull complete flow, normalized double, coefficient bundles, and distributional recurrence.
`lem:smoothing`, 273–364	Radial spatial smoothing on embedded disks at distance $r > K\varepsilon$; L^2 and scaled-gradient bounds; local commutation; for fixed u, $\varepsilon \|YS_\varepsilon u\|_2 \to 0$. Supported adjoints require $L > K + 4$.
`lem:energy`, 378–423	For compactly supported w with $Xw, Yw \in L^2$ and $Xw = 0$ off E, $\|\nabla w_j\|_2^2 \le 4 \|Xw\|_2 \|\mathbf{1}_E \nabla w\|_2$. Derivative summability belongs to the regularized field.
`lem:bounded-modes`, 425–466	For $\|h\|_\infty \le H$ and local $Xh = 0$, cone cutoff and smoothing give $\|\nabla h_j\|_2 \le CH$ for every j, not summability in j.
`prop:rigidity`, 476–676	Bounded flow invariance is claimed to imply $\partial f_j = \bar\partial f_j = 0$ for every j via two successive limits. This is the substantive low-regularity bridge.
`sec:conclusion`, 678–710	Holonomy eliminates nonzero parallel modes at an irrational vertex; connectedness fixes the zero mode.
`cor:quantum`, 712–817	Imported interior QE, boundary QE and nodal growth, conditional on the classical theorem; no rate, QUE, or mixing.

For a fixed $m \in \mathbb{Z}$ the source forms

$$U = \sum_{k \ge 0} P_{m+2k} f, \quad B = e^{i(m-1)\theta} \bar\partial f_m, \quad XU = B. \tag{9}$$

Finite tails leave a terminal mode with bounded L^2 norm and angular index tending to infinity. It vanishes weakly, not necessarily in norm. For fixed $s > 0$ let $T = s/\varepsilon$ and retain length-T segments with clearance $L\varepsilon$. A time mesh and cone area $O(\varepsilon^2)$ bound discarded mass by $C(T\varepsilon + \varepsilon^2)$. Negative-time averaging produces g_ε with

$$\|g_\varepsilon - f\|_2 \le CH\sqrt{s + \varepsilon^2}, \quad \|Xg_\varepsilon\|_2 \le 2H/T, \quad g_\varepsilon = 0 \text{ where } r < L\varepsilon. \tag{10}$$

No derivative of the good-set indicator is used: differentiation of the time average is an endpoint difference. The source integrates a spatially smoothed endpoint equation along good segments to obtain $\langle B, Yw_\varepsilon \rangle = o(1)$ for $w_\varepsilon = S_\varepsilon g_\varepsilon$. The scaled derivative convergence is used for the *fixed* U.

At fixed s, coefficient gradients are bounded by $O(H/\sqrt{s})$. The first weak limit is a bounded $g^{(s)}$ with local $Xg^{(s)} = 0$, $\left\|g^{(s)} - f\right\|_2 \le CH\sqrt{s}$, and $\left\langle B, P_{m-1} Y g^{(s)} \right\rangle = 0$ using just modes $m, m-2$. Before $s \downarrow 0$, the source reapplies its bounded-coefficient lemma to this exact local solution, obtaining a new uniform $O(H)$ bound. Strong L^2 convergence and weak convergence of those two gradients then give $\langle B, P_{m-1} Yf \rangle = 0$. Finally

$$P_{m-1} Yf = -2iB, \qquad \langle B, -2iB \rangle = 2i \|B\|_2^2$$

forces $B = 0$. Swapping the limits or treating $O(s^{-1/2})$ as uniform would not reproduce the proof.

The independent bounded audit specifically checked the supported zero-extension margin, global primitive for Stokes, regularized-only summability, weak terminal-mode limit, time-average differentiation, and the fresh bound before the second limit. Almost-every-angle smooth representatives are handled by countable charts and rational-time pullbacks, so the source does not require angular smoothing along every exceptional direction. No decisive gap was found within this scope. That outcome is a bounded analytic assessment, not a formal verification of the global theorem. Our independently proved finite-form-domain results below do not silently replace this chain by an $H^{1,0}$ assumption on indicators.

4 Holonomy: Exact and Quantitative

Proposition 4.1 (Parallel-mode obstruction)**.** *If $u \in L^2(M, L_j)$ has zero weak flat gradient, it is parallel. If $j \neq 0$ and α/π is irrational, then $u = 0$. For $j = 0$ it is constant.*

Proof. On every connected regular chart a distribution with both first derivatives zero is constant. Those constants match by the unitary transitions. Developing the punctured cone shows vector holonomy $2\pi - 2\alpha$, equivalently -2α modulo 2π; reversing orientation reverses the sign. Coefficient continuation has phase $e^{2ij\alpha}$ in the source convention. Thus $(e^{2ij\alpha} - 1)u = 0$. This multiplier is nonzero for every nonzero integer j when α/π is irrational. For $j = 0$ connectedness gives the assertion. □

Consequently, *if the source rigidity proposition is valid*, its ergodicity conclusion follows by lifting an invariant indicator and using Fourier completeness. Holonomy alone does not make arbitrary X-invariant coefficients parallel.

4.1 Finite consistency residual

For $N \geq 1$ let $\mathcal{J}_N = \{-N, \dots, -1, 1, \dots, N\}$ and

$$d_j(\alpha) = |e^{2ij\alpha} - 1| = 2|\sin(j\alpha)|, \qquad \delta_N = \min_{j \in \mathcal{J}_N} d_j(\alpha). \tag{11}$$

These are holonomy consistency numbers, not dynamical spectral gaps.

Proposition 4.2 (Finite holonomy certificate)**.** *For $c \in \mathbb{C}^{2N}$ and $r_j = (e^{2ij\alpha} - 1)c_j$, if $\delta_N > 0$, then $\|c\|_{\ell^2} \leq \delta_N^{-1} \|r\|_{\ell^2}$. If $|\alpha - \widehat{\alpha}| \leq \eta$ and $\underline{d}_j$ are rigorous lower enclosures of $d_j(\widehat{\alpha})$, then*

$$\underline{\delta}_N = \min_{1 \leq j \leq N} \max\{0, \underline{d}_j - 2j\eta\} \leq \delta_N. \tag{12}$$

For p tested loops with angles $\alpha_1, \dots, \alpha_p$, the singular value of the stacked residual map is $\min_{1 \leq j \leq N} (\sum_{\ell=1}^{p} d_j(\alpha_\ell)^2)^{1/2}$.

Proof. Square and sum the scalar residual identities. Perturbation uses $|e^{ix} - e^{iy}| \leq |x - y|$ and the reverse triangle inequality. Stacking loop residuals adds squared moduli at each mode. □

This certifies amplitudes already identified in one fiber or one parallel local model, not arbitrary spatial sections from one loop measurement. Failure of a positive lower bound is inconclusive. Finite angle intervals may exclude finitely many resonances but cannot prove irrationality of their unknown exact value.

4.2 Annular control without a curve-trace assumption

Fix an embedded cone annulus $\mathcal{A} = \{\rho < r < R\}$ at the α vertex, with $0 < \rho < R$. Its developed wedge has $(r, \varphi) \in (\rho, R) \times (0, 2\alpha)$ and metric $dr^2 + r^2 d\varphi^2$. In a parallel frame a section of L_j has twist

$$u(r, 2\alpha) = e^{-2ij\alpha} u(r, 0) \tag{13}$$

in the Sobolev trace sense. The inverse phase corresponds to the opposite cut convention and gives the same constants. Define

$$\kappa_j = \frac{\pi}{\alpha} \operatorname{dist}(j\alpha/\pi, \mathbb{Z}). \tag{14}$$

Theorem 4.3 (Twisted annular control)**.** *For every H^1 section on $\mathcal{A}$ satisfying (13), if $\kappa_j > 0$, then*

$$\|u\|^2_{L^2(\mathcal{A}, dm)} \le \frac{R^2}{\kappa_j^2} \|\nabla u\|^2_{L^2(\mathcal{A}, dm)}. \tag{15}$$

No radial boundary condition at ρ or R is needed.

Proof. For almost every r, Sobolev slicing gives an H^1 angular function. Multiplication by $e^{ij\varphi}$ makes it 2α-periodic. The twisted frequencies are $\pi n/\alpha - j$, $n \in \mathbb{Z}$, whose smallest absolute value is κ_j. Angular Parseval gives

$$\int_0^{2\alpha} |u|^2 \, d\varphi \le \kappa_j^{-2} \int_0^{2\alpha} |\partial_\varphi u|^2 \, d\varphi.$$

Multiply by r, integrate, and use $r \le R^2/r$. The resulting angular energy is part of the full gradient energy; the radial part is nonnegative. Area normalization cancels. Approximation gives the same result on the full twisted H^1 space. □

The angular constant is sharp for the one-dimensional twisted problem. The R^2 comparison is not claimed sharp on the two-dimensional annulus. Crucially, the proof uses an area family of almost-everywhere slices, not a trace of arbitrary bulk L^2 data onto a chosen loop.

4.3 From annuli to coefficient Rayleigh gaps

Assume a scalar Poincare constant $C_P > 0$ on the completed double is supplied or separately certified in the physical metric:

$$\|v - \langle v \rangle\|_2^2 \le C_P \|\nabla v\|_2^2, \qquad \langle v \rangle = \int_M v \, dm. \tag{16}$$

Such a constant exists on a fixed compact connected finite-cone surface by the usual compact-embedding and Poincare argument. A numerical value is not inferred from a sampled mesh here. Put $a = m(\mathcal{A}) > 0$.

Theorem 4.4 (Anchored coefficient gap)**.** *For $u \in \mathcal{V}_j$ and $\kappa_j > 0$,*

$$\|u\|_2^2 \le C_j q_j(u), \qquad C_j = C_P + \frac{1}{a} \left(\sqrt{C_P} + \frac{R}{\kappa_j} \right)^2. \tag{17}$$

Thus $\lambda_j = \inf_{0 \ne u \in \mathcal{V}_j} q_j(u) / \|u\|_2^2$ satisfies $\lambda_j \ge C_j^{-1}$.

Proof. The modulus $v = |u|$ is a global scalar since transitions are unitary. The weak diamagnetic inequality $|\nabla|u|| \le |\nabla u|$ follows by differentiating $(|u|^2 + t^2)^{1/2}$ locally, approximating, and letting $t \downarrow 0$. For $u \in \mathcal{V}_j$ the same approximation places v in scalar H^1 on the completion. Let $b = \langle v \rangle$ and $E = q_j(u)$. Then $\|v - b\|_2 \le \sqrt{C_P E}$ and

$$\sqrt{a}\, b \le \|u\|_{L^2(\mathcal{A})} + \sqrt{C_P E} \le (R/\kappa_j + \sqrt{C_P})\sqrt{E}.$$

Since $m(M) = 1$, orthogonality to constants gives $\|u\|_2^2 = \|v - b\|_2^2 + b^2$. Substitution proves the claim. □

Every $\mathcal{V}_j$ section restricts to the twisted H^1 annulus used above. There is therefore no change of operator domain in this application. If several cone annuli are available, take the smallest valid C_j among those with nonzero twist. The mean-free zero-mode gap is at least C_P^{-1}. At an irrational vertex,

$$\lambda_N = \min\{C_P^{-1}, C_j^{-1} : 1 \le |j| \le N\} > 0 \tag{18}$$

is a conditional explicit gap bound for a finite collection of coefficient forms. Its certified inputs are C_P, embedded annulus geometry, physical area, and positive lower bounds for the κ_j. No uniformity over degenerating triangles is supplied.

5 Finite-Band Transport and Tail Budgets

This estimate is independent of the bounded-invariant chain and applies to a declared Friedrichs form model. Its matrix algebra is classical.

Lemma 5.1 (Parity summation matrix)**.** *For the $L \times L$ upper triangular all-ones matrix T_L,*

$$\|T_L\|_{2\to 2} = \frac{1}{2\sin(\pi/(4L+2))} \le L, \qquad L \ge 1.$$

Proof. Reversing order replaces T_L by the lower triangular all-ones matrix, without changing its norm. Its inverse is the first-difference matrix D, with diagonal one and subdiagonal minus one. The matrix D^*D has diagonal $2, \dots, 2, 1$ and adjacent entries minus one. Its eigenvectors have components $\sin(j\vartheta_k)$, $1 \le j \le L$, where the endpoint conditions $x_0 = 0$, $x_{L+1} = x_L$ give $\vartheta_k = (2k-1)\pi/(2L+1)$, $1 \le k \le L$. The eigenvalues are $2 - 2\cos\vartheta_k$. The reciprocal square root of the smallest is the stated norm. The bound by L also follows from $\|T_L\|_2 \le \sqrt{\|T_L\|_1 \, \|T_L\|_\infty} \le L$. □

Theorem 5.2 (Finite-band coercivity)**.** *Let $N \ge 0$, $f = \sum_{|j|\le N} f_j e^{ij\theta}$ with $f_j \in \mathcal{V}_j$. Suppose $\lambda_N > 0$ bounds below all relevant coefficient Rayleigh gaps, with the mean removed only in mode zero. Put $K_N = \csc(\pi/(4N+6))$. Then*

$$\sum_{|j|\le N} q_j(f_j) \le K_N^2 \, \|Xf\|_2^2, \qquad \|f - \langle f \rangle\|_2 \le \frac{K_N}{\sqrt{\lambda_N}} \, \|Xf\|_2 . \tag{19}$$

The full residual includes angular modes $\pm(N+1)$.

Proof. Set $a_j = \|\partial f_j\|_2$ and $b_k = \|(Xf)_k\|_2$, extending coefficient sections by zero outside the band. The recurrence and (8) imply $a_j \le b_{j+1} + a_{j+2}$. Iteration from the upper endpoint yields $a_j \le \sum_{\ell \ge 0} b_{j+2\ell+1}$. Each parity chain has at most $N+1$ entries. Its a-vector is entrywise dominated by $T_L b$, possibly after zero padding. The two parity residual lists use disjoint Fourier indices. Lemma 5.1 and $q_j = 4a_j^2$ prove the first inequality. Apply the Rayleigh bounds to f_j for $j \ne 0$ and $f_0 - \langle f_0 \rangle$ for $j = 0$, then sum by Parseval for the second. All identities extend from the compact core by form convergence. □

For the triangle, (18) is an admissible λ_N. The constant improves the rough $2(N+1)$ allowance; their ratio tends to $2/\pi$. Here K_N is twice the exact norm of T_{N+1}, not the claimed optimal geometric transport constant. For $N = 0$ the two output modes remain present, though not both are needed in the one-sided proof. Removing them would make compressed transport zero.

Corollary 5.3 (Full-state residual budget)**.** *Let $f \in L^2(SM)$ with $Xf \in L^2$ distributionally and let $g = P_{\le N} f$ have the form-domain properties above. Suppose $\|f - g\|_2 \le \varepsilon_N$ and $\|X(f-g)\|_2 \le \tau_N$ in the physical norm. Then*

$$\|f - \langle f \rangle\|_2 \le \varepsilon_N + \frac{K_N}{\sqrt{\lambda_N}}(\|Xf\|_2 + \tau_N). \tag{20}$$

Proof. The angular projection retains the zero coefficient, so $\langle g \rangle = \langle f \rangle$. Apply the triangle inequality and $Xg = Xf - X(f-g)$ to (19). □

The derivative tail is necessary: X and angular projection do not commute. An angular H^s prior gives $\varepsilon_N \le (N+1)^{-s} \||\Theta|^s f\|_2$ by Parseval, but does not give τ_N without horizontal-derivative control. Invariant indicators supply neither prior. To force constancy for an invariant f using this budget would require a sequence with $\varepsilon_N \to 0$ and $K_N \tau_N / \sqrt{\lambda_N} \to 0$. Minimizing the certified right side over finitely many admissible N is legitimate adaptation, not an unmeasured computational speedup.

5.1 Small divisors and the absence of a rate claim

For irrational $x = \alpha/\pi$, the pigeonhole principle applied to $0, x, \dots, Qx$ modulo one gives $1 \le q \le Q$ with $\operatorname{dist}(qx, \mathbb{Z}) \le 1/Q$. Such q are unbounded; otherwise a finite set of nonzero distances would have positive minimum. Thus $d_q(\alpha) \le 2\pi \operatorname{dist}(qx, \mathbb{Z}) \to 0$ along a subsequence and no positive lower bound for δ_N survives as $N \to \infty$. This proves the failure of this holonomy-based uniform constant, not a characterization of the full dynamical spectrum. Neither (19) nor (20) is an ergodic or mixing-rate estimate. Forni–Moll's Diophantine estimates [2, Theorem 4.3] concern a horizontal operator under additional arithmetic assumptions, not an unconditional rate for billiard flow.

6 Ram Representation and Physical Residuals

6.1 What the actual private passages supply

The private polygon report [12], in its section on ergodicity and the scope of Ram methods, distinguishes rational directional dynamics from non-rational full flow. Its Cheeger program requires uniformity in shrinking singular neighborhoods and deformation scales. Its parameter-genericity target is not every irrational triangle. Finite-scale panels do not establish the source's bounded-invariant-to-parallel passage. Its physical-Gram discussion also warns that coordinate scaling preserves roots, not raw Euclidean condition numbers.

Actual bundle and covariant-moment passages in [13] prescribe source-to-target transport, measure factors, and connection laws. Their geometric averaging formula specializes on an embedded flat disk to

$$(S_\varepsilon u_j)(x) = \int \rho_\varepsilon(y) \mathcal{P}_{x-y \to x} u_j(x-y)\, dy. \tag{21}$$

The disk avoids cone tips and the transport path is its short straight segment. In a parallel frame $\mathcal{P}$ is the identity and the Jacobian is one. In different frames the kernel transforms at both

endpoints. This is the precise Ram geometric specialization of the source smoothing, not a new global convolution on a cone surface.

For $p+1$ bounded covariant derivatives throughout that disk, Taylor's formula yields

$$S_\varepsilon u_j = \sum_{|a|\le p} \frac{(-1)^{|a|} m_a(\varepsilon)}{a!} \nabla^a u_j + R_{p+1}, \qquad m_a(\varepsilon) = \int \rho_\varepsilon(y) y^a \, dy. \tag{22}$$

If C_{p+1} bounds covariant directional derivatives of order $p+1$, then

$$|R_{p+1}| \le \frac{C_{p+1}}{(p+1)!} \int \rho_\varepsilon(y) |y|^{p+1} \, dy.$$

Odd moments vanish for a radial kernel. This is exact on the declared polynomial class and remainder-controlled on the declared smooth class, not a Taylor expansion of an arbitrary bounded invariant.

The physical-domain chapter [14] requires transport of $\mathrm{Dom}(A)$ as well as A. Accordingly, (6) transports the generator to $\mathcal{F}\mathsf{X}\mathcal{F}^{-1}$ on $\mathcal{F}\,\mathrm{Dom}(\mathsf{X})$; the recurrence describes its expression, not a freely chosen domain on coefficient sequences. The original private chapter uses conjugate-first inner products; only the explicit convention changes here, not its norm or domain conclusions.

6.2 A conforming finite model

Choose smooth local section atoms with compact regular support and assemble them by (4). Examples are windowed polynomial or exponential-polynomial atoms $\chi(x) e^{s\cdot x} x^a$ in each mode, including cutoff derivatives and transition factors. Let $J : \mathbb{C}^d \to L^2(SM)$ be injective synthesis with columns ψ_p. Then

$$G_{pq} = \int \overline{\psi_p} \psi_q \, d\nu, \qquad \|Jc\|_2^2 = c^* G c, \qquad P = J G^{-1} J^*.$$

Here J^* is the Hilbert adjoint; in coordinates it consists of the coefficient integrals $\int \overline{\psi_p} u \, d\nu$. Set $K_{pq} = \int \overline{\psi_p} X \psi_q \, d\nu$ and $B = G^{-1} K$. Orthogonal projection gives

$$\|XJc\|_2^2 = c^* B^* G B c + \|(I-P) X J c\|_2^2 . \tag{23}$$

Indeed $PXJc = JBc$, and the two components are orthogonal. This specializes the residual decomposition in the actual private physical-metric chapter [15]. Its first term alone is not an upper bound on the full residual.

A certificate must therefore carry the atoms and twists, a positive physical Gram enclosure, the full transport image (outer angular modes and spatial leakage included), enclosed angle and geometry constants, and tail bounds for any state outside the model. Orthogonalization by $G^{1/2}$ changes coordinates but does not eliminate leakage. A sampled matrix gap is not continuum ergodicity. No software speedup is measured or asserted.

7 Conditional Finite-Cone Extension

The source analytic chain visibly uses finite-cone geometry rather than the number three. This observation can be stated without claiming an independently settled general billiard theorem.

Theorem 7.1 (Source-dependent transfer)**.** *Let $\overline{M}$ be a compact connected oriented Euclidean cone surface with finitely many cone points of finite positive angles and positive finite area; write M for its regular part. Assume its unit geodesic flow is almost everywhere complete and measure preserving. Assume also the following* analytic transfer hypothesis*: the supported smoothing, localized energy, bounded-coefficient, and two-successive-limit arguments of source* `lem:smoothing`*,* `lem:energy`*,* `lem:bounded-modes`*, and* `prop:rigidity` *are valid on this M with constants depending on the fixed surface but not the angular mode or smoothing scale. If the rotational holonomy image in $SO(2)$ is infinite, the geodesic flow is ergodic.*

Proof. The analytic transfer hypothesis gives parallel angular coefficients for each bounded invariant function. An infinite subgroup of the circle is dense: a proper closed subgroup has a smallest positive rotation and is finite. For each $j \neq 0$ the kernel of the character $e^{i\phi} \mapsto e^{-ij\phi}$ is finite. Therefore that character cannot be identically one on the infinite holonomy image, and a parallel L_j section must vanish. The zero coefficient is constant by connectedness. Fourier completeness makes every bounded invariant constant; applying this to invariant indicators proves ergodicity. □

The transfer hypothesis is substantive and source-dependent, not established by the elementary holonomy proof. Its geometric inputs are: embedded flat disks at distance $K\varepsilon$ from tips; tip area $O(\varepsilon^2)$; cutoffs with $O(\varepsilon^{-1})$ gradients; finite total area; and the stated conull complete measure-preserving flow. Flat oriented transitions supply unitary bundle algebra and local commutation. The analytic route then repeats the endpoint, long-segment, and refreshed two-limit estimates in Section 3. Genus zero, exactly three vertices, or convexity of the polygon are not explicit inputs of these estimates. This supports a conditional extension; it is not a new unconditional proof of every irrational polygon's ergodicity.

For example, the double of a finite simple Euclidean polygon, allowing reflex angles in $(0, 2\pi)$, has finitely many positive cone angles. One irrational polygon angle supplies infinite rotational holonomy. The geometric hypotheses fit, but an assertion of ergodicity by this route remains conditional on the full analytic transfer hypothesis. In contrast, finite holonomy preserves a nonconstant measurable direction class modulo its finite group, hence full geodesic flow is not ergodic. This does not contradict directional ergodicity.

The codimension of the singular geometry matters. If a different setting had singular-neighborhood volume $O(\varepsilon^{\varkappa})$ with the same $O(\varepsilon^{-1})$ cutoff cost, the product of the two localized derivative norms would scale as $\varepsilon^{\varkappa-2}$ and the mesh loss as $T\varepsilon^{\varkappa-1} + \varepsilon^{\varkappa}$. The exponent $\varkappa = 2$ is critical for the specific balance used by the source. This scaling calculation is not a theorem for other singular flows.

8 Quantum and Nodal Consequences

8.1 Realizations and quantifiers

Fix a triangle and *assume its classical billiard flow is ergodic*, either by an established special-case theorem or conditionally from the pinned source. Dirichlet and Neumann positive Laplacians are the self-adjoint form realizations of $\int_Q |\nabla u|^2 dA$ on $H_0^1(Q)$ and $H^1(Q)$. For either condition B, choose any complete orthonormal eigenbasis with multiplicities,

$$-\Delta_B u_k = \omega_k^2 u_k, \qquad 0 \leq \omega_1 \leq \omega_2 \leq \cdots .$$

The finite-dimensional zero-frequency space does not affect density. A set $S \subset \mathbb{N}$ has density one if $\#(S \cap \{1, \ldots, K\})/K \to 1$.

Embed $\overline{Q}$ isometrically in a sufficiently large flat torus. The boundary is Lipschitz; its open sides lie in embedded smooth lines with the interior locally on one side. Its vertices form a finite codimension-two singular set; incident lines meet transversely. The ambient metric is a smooth extension. These observations check the geometric hypotheses of the imported piecewise-smooth results [6, 7, 8, 9]; the boundary is not falsely treated as globally smooth.

Interior QE supplies a density-one subsequence with

$$\langle Au_k, u_k\rangle_{L^2(Q)} \longrightarrow \int_{Q\times S^1} \sigma_0(A)\, d\mu \tag{24}$$

for every order-zero classical pseudodifferential operator having compact kernel support in $Q \times Q$ [6, 7]. A countable dense-family argument and norm bounds give a common subsequence for the observable class. It may depend on the table, boundary condition, and basis. Zero-density exceptional subsequences are not excluded.

8.2 Collision measure and boundary weights

Let Γ be the union of open sides and $B^*\Gamma = \{(s,\eta) : s \in \Gamma, |\eta| < 1\}$. Here s is arc length and η a signed tangential covector. For outgoing angle φ from the inward normal, $\eta = \sin\varphi$, so $\cos\varphi\, ds\, d\varphi = ds\, d\eta$. The collision map preserves this measure after removing vertex hits and tangencies.

Flight coordinates express flow measure as proportional to $ds\, d\eta\, dt$ for $0 \le t < \tau(s,\eta)$, with $0 < \tau \le \operatorname{diam} Q$ almost everywhere. A nontrivial invariant collision set would suspend to a nontrivial invariant flow set: both parts have positive integral of τ because $\tau > 0$ almost everywhere. A uniform positive lower bound for the roof is unnecessary. Thus flow ergodicity implies the collision-map ergodicity required by boundary QE.

Put $h_k = \omega_k^{-1}$ for $\omega_k > 0$ and define

$$b_k^D = h_k \partial_\nu u_k|_\Gamma, \qquad b_k^N = u_k|_\Gamma. \tag{25}$$

Hassell–Zelditch [8, Theorem 1.1] then gives, on a density-one subsequence, for $a \in C_c^\infty(T^*\Gamma)$ and properly supported order-zero semiclassical quantization,

$$\left\langle \operatorname{Op}_{h_k}(a) b_k^B, b_k^B \right\rangle_{L^2(\Gamma)} \longrightarrow \frac{2}{\pi A_Q} \int_{B^*\Gamma} a(s,\eta) w_B(\eta)\, ds\, d\eta, \tag{26}$$

where

$$w_D(\eta) = \sqrt{1-\eta^2}, \qquad w_N(\eta) = (1-\eta^2)^{-1/2}.$$

The prefactor is $4/(\operatorname{vol}(S^1)A_Q) = 2/(\pi A_Q)$. The source correctly absorbs the original theorem's ω_k^{-2} factor into Dirichlet data. Normal reversal changes their sign, not the quadratic pairing. The data have no further unit-boundary-norm normalization.

The quantum boundary weights differ from collision measure. For a regular-boundary cutoff $\chi(s)$ and tangential symbol equal to one on $[-1,1]$, direct integration yields

$$\begin{aligned} \frac{2}{\pi A_Q} \int \chi\, ds \int_{-1}^{1} w_D\, d\eta &= \frac{\int \chi\, ds}{A_Q}, \\ \frac{2}{\pi A_Q} \int \chi\, ds \int_{-1}^{1} w_N\, d\eta &= \frac{2\int \chi\, ds}{A_Q}. \end{aligned} \tag{27}$$

The angular integrals are $\pi/2$ and π. The Neumann glancing singularity is integrable, not bounded. Corner observables require the imported nonconcentration analysis, not an arbitrary extension of compact regular-boundary test symbols.

8.3 Nodal conclusion

For a real-valued eigenbasis, Hezari's theorem [9, Theorem 1.6 and Definition 1.16] gives a density-one subsequence on which the number of connected components of $Q \setminus \{u_k = 0\}$ tends to infinity. The piecewise-smooth geometric conditions are checked above. This imports restriction, supremum and nodal arguments; holonomy algebra alone does not prove it. A sign-domain conclusion is not applied to arbitrary complex eigenfunctions. Intersecting the interior, boundary and nodal density-one sets preserves density one. No nodal rate, QUE, small-scale equidistribution, or spectral-statistics law follows.

9 Examples and Verification

9.1 Rational and irrational controls

For an equilateral triangle, $\alpha = \pi/3$ and $d_3(\alpha) = 0$. Mode three is not eliminated by rotational holonomy. With a side horizontal, $\cos(6\theta)$ is preserved by each side reflection and by straight flight; its level sets witness nonergodicity of full rational flow. This is consistent with almost-every-direction unique ergodicity on an unfolded translation surface.

For an irrational right triangle take angle ratios

$$x = \frac{\sqrt{2}}{4}, \qquad \frac{1}{2}, \qquad \frac{1}{2} - x. \tag{28}$$

They are positive, and the first and third are irrational. Among $1 \le j \le 12$, the smallest $\operatorname{dist}(jx, \mathbb{Z})$ occurs at $j = 3$, with value $3\sqrt{2}/4 - 1$. Hence

$$\kappa_{\min,12} = \frac{3x - 1}{x} = 3 - 2\sqrt{2} > 0.$$

Using $2\sin(\pi d) \ge 4d$ for $0 \le d \le 1/2$ gives the rigorous lower bound $\delta_{12} \ge 3\sqrt{2} - 4$. The exact sine value is larger. The evidence script encloses x by rational endpoints whose squares straddle $1/8$, and verifies the minimum with exact integer-distance intervals. The near-resonance sequence $j = 3, 17, 99, 577$ has distances decreasing from about $6.1 \cdot 10^{-2}$ to $3.1 \cdot 10^{-4}$. These decimal illustrations are not used to certify irrationality or ergodicity.

A geometric realization is the right triangle with vertices $(0,0), (1,0), (0, \tan(\pi x))$. The angle at $(1,0)$ is πx. Choose $R = 1/8$ and $\rho = R/2$ there. This cone annulus in the double is embedded, since the opposite side and other vertices are farther away than R. Its normalized area is

$$a = \frac{\pi x(R^2 - \rho^2)}{\tan(\pi x)}.$$

Substitution into (17) with $3 - 2\sqrt{2}$ gives a conditional finite-12 gap bound as a function of supplied C_P. No numerically certified value of C_P for this triangle is reported.

9.2 Manufactured tests and exact failure controls

The 178 author checks passed. They verify reflection products and phase composition, rational resonance, algebraic near-resonance enclosures, twisted frequencies, boundary-weight integrals, and finite parity-matrix identities. Exact arithmetic is distinguished from floating diagnostic eigenvalue and norm comparisons. The independent matrix check reports 256 assertions at sizes 1–64; its floating discrepancies do not validate cone geometry, Rayleigh constants or ergodicity.

An exact full-residual failure control is $N = 0$. If u is a nonconstant smooth compactly supported scalar and $f(x, \theta) = u(x)$, then $P_{\le 0} X f = 0$ but angular averaging gives

$$\|Xf\|_2^2 = \tfrac{1}{2} \|\nabla u\|_2^2 > 0.$$

Thus compressed zero residual may miss all transport energy.

There is also a concrete reason to require projected form regularity. On the flat 2π-periodic torus, take a smooth ϕ supported in $(-1, 1)$ with nonzero integral, and near $\theta = \pi/2$ put $g_k(\theta) = \sqrt{k}\phi(k(\theta - \pi/2))$. The field

$$f(x, y, \theta) = \sum_{k \ge 1} k^{-1} e^{ikx} g_k(\theta)$$

belongs to $\mathrm{Dom}(\mathsf{X})$ for $X = \cos\theta\, \partial_x + \sin\theta\, \partial_y$. Indeed spatial orthogonality and $\|g_k\|_2 = O(1)$ give square-summable L^2 terms; on their supports $|\cos\theta| \le 1/k$, so the squared X norms are $O(k^{-2})$ as well. Partial sums converge in the graph norm. But the zero angular coefficient has spatial Fourier coefficients $ck^{-3/2}$ with fixed $c \neq 0$. Its horizontal energy contains $|c|^2 \sum k^{-1} = \infty$, so $P_0 f \notin \mathcal{V}_0$. This is a domain counterexample on a torus, not an invariant function or an irrational-triangle counterexample. There is no orbit simulation, computed billiard eigenbasis, or empirical mixing fit in this report.

10 Literature, Benefits, and Limits

10.1 Current primary-source comparison

Primary material was checked on October 8, 2026. Kerckhoff–Masur–Smillie [3] provides the rational directional framework and generic polygonal conclusions. Vorobets [4] gives a very fast approximation criterion. Neither asserts every irrational triangle. Chaika–Forni's published 2026 result [5] gives weak mixing on a dense G_δ set of polygons, not every table or full parameter measure; its status and quantifier were checked at the journal.

Forni–Moll [2] proves ergodicity of the two-dimensional holonomy foliation under non-rational holonomy. Its Corollary 4.2 concerns $H^{1,0}$ invariant functions. Both horizontal directions, or the Sobolev assumption, matter. It cannot be directly applied to an arbitrary bounded invariant indicator. The source's claimed advance is precisely the missing low-regularity passage.

Numerical primary studies of irrational and symmetric triangles [10, 11] investigate finite trajectories and apparent localization. They do not produce the positive-measure invariant set needed to disprove the source claim; conversely apparent numerical equilibration would not prove it. A public search also returned a third-party index of the source collection, which was not used as independent corroboration. This targeted check is not an exhaustive historical novelty search.

10.2 Advantages, costs, and useful next steps

The representations expose chart twists, domains, physical norms and missing residual components. Annular coercivity turns angular monodromy into a bulk energy bound without an invalid curve trace. Anchored Poincare control gives a declared coefficient gap and the finite-band budget states exactly which derivative tail is needed. These are local verification improvements, not a replacement for the source's global analytic content.

Small divisors worsen conditioning; singular neighborhoods require correct cutoff geometry. A physical Gram matrix and leakage budget cost more than an isolated compressed matrix. Numerical constants need certified C_P and geometry. No uniformity is supplied as angles degenerate or the mode cutoff tends to infinity.

Concrete future work is to certify a scalar Poincare bound for a fixed double, enclose annulus geometry on an angle box, compare multi-cone and single-cone bounds, and construct conforming section bases with both full transport residuals and derivative tails controlled. An upgrade of (20) needs an explicit class with sufficiently small τ_N, not merely angular smoothness. Independent expert scrutiny of the source analytic chain remains necessary before promoting its global claim.

11 Assessment and Reproducibility

The realized contribution combines Proposition 4.2, Theorems 4.3, 4.4, 5.2, and Corollary 5.3, with a carefully source-dependent geometric extension. The underlying ingredients are classical. No historical novelty, new universal proof, or speedup is claimed. C2 is a conservative internal rating for a useful but limited verification improvement.

Dedicated evidence contains read scopes and SHA-256 hashes, public URLs and theorem locations, exact-check code and results, native compiler outcome, physical export logs and all-page visual inspection. No private content was used in public queries.

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